

Bees & Vibrations.

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Introduction. Firstly, thank you, everyone, for your patience and perseverance last Friday when there was a problem accessing the latest blog. The problems should be fully resolved now. I gave my “Observations at The Hive Entrance” presentation to the Welsh Beekeepers’ Association last week via Go To Webinar. It was a real privilege to be invited back to speak to them again, and they are a great group with lots of good questions at the end which I learned from too. I’ve just been out to photograph my hive entrances to show how I reduce them according to the time of year and what’s going on. I don’t go by any hard and fast rules or exact measurements, but I do keep the entrances reduced all year round which more in keeping with how bees in the wild would live. Reduced entrances makes it a lot easier for guard bees to defend the nest at certain times of the year when robbing is a risk and wasps are on the hunt for food. Listening to the winter cluster. Someone asked if I’ve ever heard the winter cluster using a stethoscope to assess a colony’s state of survival. This question sparked off a literature search and I’ve found some pretty interesting research on honey bees and vibrations to share with you. I have tried to listen to the winter cluster using a stethoscope I had for work, but I couldn’t hear a thing so I gave it away to a charity who needed it more than me. However, I think that the key to hearing bees in their winter cluster is to give a gentle tap in the side of the hive and listen with your ear flush with the brood box. This works for me every time now but if I had tapped the hive while using the stethoscope it would have worked too, I think. Startle reflex. I was asked to describe the noise that I heard with my ear flat against the brood box, and it sounded to me like gentle rustling or murmuring voices in the distance. It was like a quick reply to my tap and it faded away quickly too. Scientists¹ describe it aptly as a hissing sound. They relate this reflex response to the human newborn baby’s startle reflex. The startle reflex is used formally in the newborn Moro reflex arc neurological test to assess normal neuromuscular functioning. If you have ever held a newborn baby, you will remember the startle reflex and how the little one threw its hands out in alarm when he or she was moved suddenly. Although the reflex arc has not been described in invertebrates, the researchers likened the response of the colony to vibrations to human babies being startled. They interpreted the rapid response as a danger signal to bees. It is a neurological response involving motor neurones. Virgin piping. What I went on to discover is quite fascinating. I already knew that when newly emerged virgin queens start piping, workers freeze on the frame. Workers busy vibrating queen cells and preventing those queens from emerging also stop still too. All is quiet. Only the piping queen is heard, and she can hear her quacking (lower pitched piping) rivals who are about to emerge. However, I was amazed to learn that narrow- band vibrational waves can immobilise a colony for up to four hours without harming them. No ill effects were observed in the experiment that revealed this¹. In the wild. In the wild, honey

bees might become immobilised in their nests if branches are continually knocking against their tree nests. Thunder can induce vibrations apparently. I am sure now that vibrations account for the stories that I have heard of people transporting bees without shutting them in. I heard of this happening to a new beekeeper who had gone to collect a swarm in the country last year. As she was driving back towards town, things were all quiet in the back till she stopped at the first set of traffic lights. The next thing she knew the back windows were black with bees. The experiment. In the experiment¹, the scientists delivered a weak short-lived automated vibrational pulse to 9 colonies over several months. They sent this pulse at random times, approximately every hour, to prevent the colony habituating to it and stopping reacting. Humans habituate to noise and soon stop noticing it, for example, if they move from the quiet countryside to live on a busy street in a city. The reactions of the bees were measured using an accelerometer which is an instrument for measuring the accelerations of a moving or vibrating body. The accelerometers were placed in the middle of the central frame in each hive and a beep pulse of 340 Hz was delivered to stimulate the colony. The scientists measured the responses which consisted of two parts. The first part was a short-lived response of decreased vibrations resulting from the colony freezing on the spot and becoming temporarily immobile. The second part produced a relatively long-lived response of increased vibrations that suddenly tailed off. The scientists suggest that the first part demonstrates the overall mobility of the colony, and the second part correlates to the colony restfulness, or lack of stress. Only a few bees produced the whooping or begging signal² which send a couple of messages and occur when bees collide with each other. However, the most interesting discovery in this experiment was that the vibrations reaching the bees varied in amplitude across the year. The amplitude of noise response was highest in March when all the colonies were short of stores. However, it can vary across a day. Scientists found that a relatively weak vibration pulse given in mid-December got a very great response from the bees, whereas in October when they gave a stronger signal it hardly produced a response at all. So, the long-lived second part of the response is almost absent during the active season in a healthy colony when there is so much collecting and storing going on in the hive which is buzzing with activity. Sometimes in the early mornings of the active season the researchers elicited a second-phase response. Conclusion. So, how can this research be useful to beekeepers? Well, it is a non-invasive way to assess a colony's overall mobility, clustering, and peacefulness. One of the nine colonies developed serious health problems and dwindled over the season. It was the only colony to demonstrate an easy-to-measure strong second long-lived phase of buzzing response throughout the summer. This colony was weak and not as busy as the others at this time. Giving vibration signals might be a way of measuring colony health in the future, but it also reveals another fascinating aspect of bee behaviour and response to vibrations. References: ¹Bencsik, M., McVeigh, A., Claeys Bouuaert, D. et al. Quantitative assessments of honeybee colony's response to an artificial vibrational pulse resulting in non-invasive measurements of colony's overall mobility and restfulness. *Sci Rep* 14, 3827 (2024). <https://doi.org/10.1038/s41598-024-54107-8> ²Ramsey M, Bencsik M, Newton MI (2017) Long-term trends in the honeybee 'whooping signal' revealed by automated detection. *PLoS ONE* 12(2): e0171162. doi:10.1371/journal.pone.0171162 Discover more from The Beelistener Subscribe to get the latest posts sent to your email. Wow! We go around our daily lives and now we know that bees resonate with different frequencies. What an amazing insect. Thank you for sharing. As a newly starting out beekeeper, I'm keen to learn. Hello Mitch. Thank you for commenting and I'm glad you enjoyed the blog. I hope you have a great first beekeeping season. Thank you Anne for another interesting blog. I tried to leave a donation through your PayPal icon but it stated that 'this account is no longer active'. Just thought you may not be aware of this. Hello Margaret. Thank for your positive comments on the blog and for letting me know about the PayPal account not working. I will send you an email. I encourage Bencsik et al. to further their studies of the bees' responses to comb vibrations by working with colonies living in observation hives.. This way, they will be able to see, as well as hear, how the bees respond to the vibrations/sounds that they (the bees and the biologists) have detected. I find that carefully observing and then describing how a worker bee behaves after receiving a signal provides

good clues about the function of this signal. Also, reporting behavioral observations will make the authors' studies "come alive" because they will be able to report what they have seen, as well as what they have heard. Hello Tom. Thank you for mentioning this aspect of research and how beneficial it would have been to have done this piece of work using an observation hive. Hi Ann, I am Saba Mustafa, a PhD student in New Zealand and currently working on my research project related to bees. I want to know more about the reaction of Bees to external vibrations, especially during extreme weather conditions or during natural disaster situation. Hello, Saba. That sounds like an interesting topic. I haven't come across any information yet relating to bees and extreme weather events etc. but it is something to look into that will involve a bit of research. Have you looked at any of Lars Chittka's work for your own project? It might be worth contacting him. He might know of someone who is working on this area of bee behaviour if he is not.